The TNT and TNT3c recordings: what is measured

> **QUESTION**
>
> What do the TNT and TNT3c recordings contain, and which numbers are taken from them for the comparison with the model?

INPUTS

- `TNT_ovlp16_resp_perfly.pkl` and `TNT3c_ovlp16_resp_perfly.pkl` (Yang, 2026-10-15): per recording, a table of units × 16 odors
- `TNT_ovlp16_cosine_mean_RDM.csv`, `TNT3c_ovlp16_cosine_mean_RDM.csv`: Yang's mean RDMs, used as a check

OUTPUTS

- `with_outside_ROI_synapses/claws_v2m2/odor_weighted/TNT_vs_model_per_odor.csv`, `TNT_vs_model_odor_pairs.csv` (data columns), `TNT_rank_order.csv` (data rows)
- `odor_weighted/TNT_data_checks.csv` / `.json` (data only)

SCRIPTS

- `scripts/tnt_compare.py` (section "data" and the helpers)
- `scripts/tnt_data_checks.py`
- `scripts/tnt_rank_order.py`
- `scripts/reports/r06_tnt_data.py` (this report)

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Yang recorded KC responses to 16 odors in two genotypes. In **TNT**, APL's synaptic output is blocked (tetanus toxin light chain expressed in APL). **TNT3c** is the control, with APL intact (Yang, 2026-10-15). This report describes the files, the quantities taken from them, and how reliable those quantities are. The pickles are read with a restricted loader (`scripts/orn_load.py`) that only accepts pandas and numpy classes.

What changed in this revision (2026-11-04)

The responses are FC z-scores (Yang, 2026-11-03); "responding" means above 1 z-score unit.

New: what the per-odor effect means mechanistically, tested against chance overlap computed from the data alone.

New: the statistics behind the per-odor effects (permutation null for the additive fit, bootstrap ranges, split-half reproducibility).

Numbers that involve the model now come from the run with the corrected calyx and the new claws.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
c	Condition: TNT (APL blocked) or TNT3c (control, APL intact)	

Symbol	What it stands for	Values, units, range
n	A recording. Each recording is one hemisphere and is treated as independent (Yang, 2026-10-15)	8 TNT, 11 TNT3c
i	A unit (one row of a recording's table): a segmented KC response region	1,638–2,462 per recording
o, a, b	Odors	16, the odors of the ORN table
$r_{io}^{(n)}$	Response of unit i in recording n to odor o, as provided	FC z-score (`fc_zscore`); −2.6 to 7.4
$\phi_o^{(n)}$	Fraction of recording n's units with $r_{io}^{(n)} > 1$ ("responding")	0 to 1
ϕ_o^c	Mean of $\phi_o^{(n)}$ over the recordings of condition c	0 to 1
$S_{ab}^{(n)}$	Cosine similarity of odors a and b over the units of recording n	−1 to 1
S_{ab}^c	Mean of $S_{ab}^{(n)}$ over the recordings of condition c (= 1 − Yang's RDM)	−1 to 1
ΔS_{ab}	Effect of blocking APL on the pair: $S_{ab}^{\mathrm{TNT}} - S_{ab}^{\mathrm{TNT3c}}$	positive = more similar when APL is blocked
s_o	Suppression of odor o by APL: $\phi_o^{\mathrm{TNT}} / \phi_o^{\mathrm{TNT3c}}$	≥ 1 means more units respond when APL is blocked
$\mu,\ e_a$	Additive fit of ΔS: mean change μ and per-odor effect e_a (written f_a in the notes)	Section 5
ρ_s	Spearman rank correlation	−1 to 1

1 The recordings

There are 8 TNT recordings from 7 flies and 11 TNT3c recordings from 8 flies. A recording name reads `date_fly` + `FL`/`FR` (left or right side) + `fun` and a number. Where one fly gives two recordings, they are its two hemispheres. The label 20240802_5 appears in both files but names two different flies (Yang). Every recording is a table with one row per unit and one column per odor; the 16 odors are the same, in the same order, as in the ORN table.

Table 2. The recordings. "values > 1" is the share of all unit × odor responses above 1, the "responding" criterion used throughout.

condition	recording	side	units	values > 1
TNT3c (control, APL intact)	`20240710_2FRfun1`	R	1,952	9%
TNT3c (control, APL intact)	`20240712_2FLfun1`	L	1,947	8%
TNT3c (control, APL intact)	`20240726_4FRfun1`	R	1,638	8%
TNT3c (control, APL intact)	`20240801_3FLfun1`	L	1,956	7%
TNT3c (control, APL intact)	`20240801_3FRfun1`	R	2,070	5%
TNT3c (control, APL intact)	`20240809_3FRfun1`	R	1,961	6%
TNT3c (control, APL intact)	`20240710_2FLfun1`	L	1,790	7%
TNT3c (control, APL intact)	`20240809_3FLfun2`	L	2,219	4%
TNT3c (control, APL intact)	`20240814_1FLfun1`	L	2,350	10%
TNT3c (control, APL intact)	`20240802_5FRfun2`	R	2,374	5%

condition	recording	side	units	values > 1
TNT3c (control, APL intact)	`20240830_3FRfun2`	R	1,894	6%
TNT (APL blocked)	`20240717_2FLfun1`	L	2,233	16%
TNT (APL blocked)	`20240717_3FLfun1`	L	2,462	19%
TNT (APL blocked)	`20240725_1FLfun1`	L	1,883	16%
TNT (APL blocked)	`20240801_4FRfun1`	R	1,728	15%
TNT (APL blocked)	`20240802_5FLfun1`	L	1,812	13%
TNT (APL blocked)	`20240807_3FLfun1`	L	2,253	14%
TNT (APL blocked)	`20240814_3FLfun1`	L	1,880	14%
TNT (APL blocked)	`20240814_3FRfun2`	R	2,238	15%

2 Response values and the "responding" criterion

The responses $r_{io}^{(n)}$ are **FC z-scores** (`fc_zscore`; Yang, 2026-11-03). Most values lie near 0, and a minority are large. A unit **responds** to an odor if its value is above 1 z-score unit. This is the threshold used in all comparisons (0.5 is checked as an alternative):

$$\phi_o^{(n)} = \frac{1}{N_n}\,\#\{\,i : r_{io}^{(n)} > 1\,\}, \qquad \phi_o^{c} = \frac{1}{|c|}\sum_{n\in c}\phi_o^{(n)} \tag{1}$$

N_n = units in recording n; $|c|$ = recordings of condition c; every recording counts equally

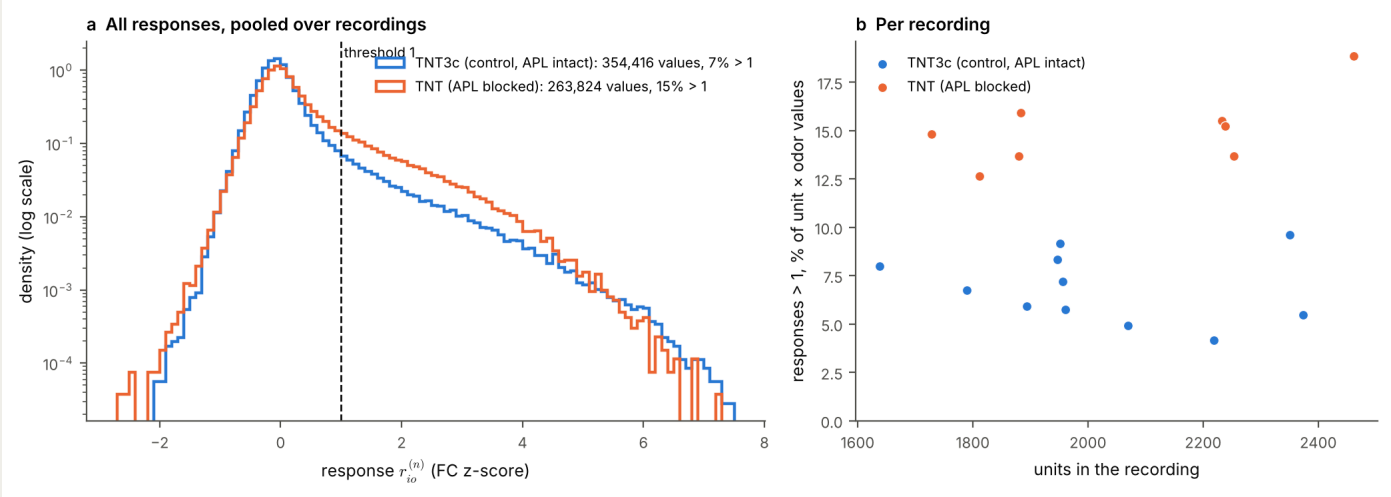


Figure 1. Response values in the two conditions. a, Distribution of every response value (unit × odor × recording) in each condition, as a density on a log scale. The dashed line is the responding threshold of 1. TNT has a heavier right tail. **b**, Each point is one recording: its number of units (horizontal) and the percentage of its unit × odor values above 1 (vertical).

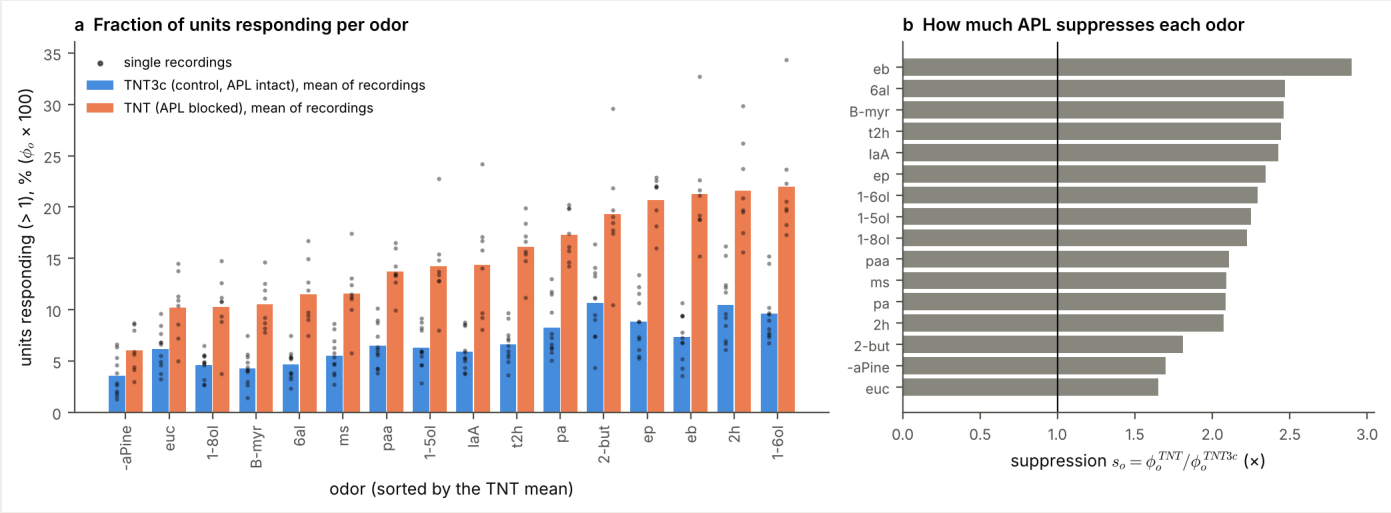


Figure 2. Fraction responding per odor, and suppression. **a**, Percentage of units responding (> 1) to each odor. Bars are the mean over recordings (blue TNT3c, orange TNT); black dots are single recordings. **b**, Suppression per odor: the TNT mean divided by the TNT3c mean. A value of 2 means twice as many units respond when APL is blocked.

$$s_o = \frac{\phi_o^{\mathrm{TNT}}}{\phi_o^{\mathrm{TNT3c}}} \tag{2}$$

On average over odors, 15% of units respond in TNT and 7% in TNT3c. Suppression ranges from ×1.65 (euc) to ×2.90 (eb). Report 07 uses these per-odor fractions to pin the model: with APL blocked, the model responds to each odor in the TNT fraction, and with APL intact its mean equals the TNT3c mean.

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3 Odor-pair similarity

For each recording, the similarity of two odors is the cosine of their response vectors over all of its units (raw values, not thresholded). The condition's similarity is the mean over its recordings:

$$S_{ab}^{(n)} = \frac{\sum_i r_{ia}^{(n)} r_{ib}^{(n)}}{\sqrt{\sum_i \left(r_{ia}^{(n)}\right)^2}\sqrt{\sum_i \left(r_{ib}^{(n)}\right)^2}}, \qquad S_{ab}^{c} = \frac{1}{|c|}\sum_{n \in c} S_{ab}^{(n)} \tag{3}$$

$1 - S_{ab}^{c}$ is exactly Yang's RDM for that condition (checked to 10^{-6} for every pair). The effect of blocking APL on a pair is

$$\Delta S_{ab} = S_{ab}^{\mathrm{TNT}} - S_{ab}^{\mathrm{TNT3c}} \tag{4}$$

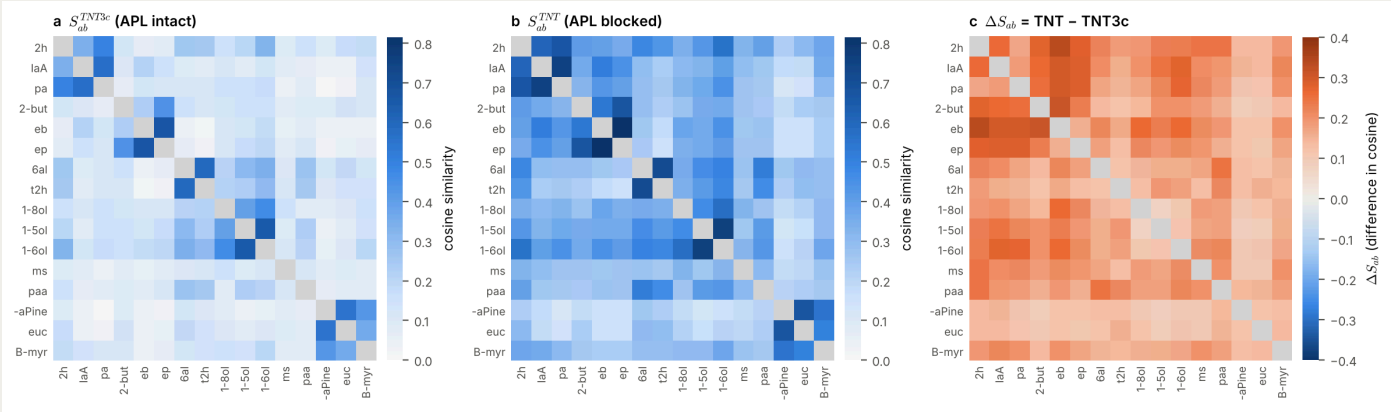


Figure 3. Odor-pair similarity in the two conditions. a, **b**, Mean cosine similarity of the units' responses for every odor pair, in TNT3c and TNT, on one scale (white 0, dark blue the highest value; slightly negative values, such as pa–-aPine in TNT3c, also show as white). **c**, The change, on a diverging scale (orange = more similar in TNT). Every pair is orange: blocking APL makes all 120 pairs more similar. Diagonal cells are left blank.

All 120 pairs are more similar in TNT. Mean similarity rises from 0.16 to 0.34 (mean ΔS = +0.178, range +0.07 to +0.33).

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4 Splitting the change into per-odor and pair-specific parts

Some odors gain similarity with every partner. To separate that from what is specific to a pair, the change is fitted by least squares as a mean plus one effect per odor:

$$\Delta S_{ab} \approx \mu + e_a + e_b, \qquad \sum_a e_a = 0 \tag{5}$$

μ = mean change; e_a = how much more the pairs of odor a change than the average pair

R^2 of this fit is 0.65: 65% of the variation of ΔS across pairs is per odor. The residuals, $\Delta S_{ab} - (\mu + e_a + e_b)$, are the pair-specific part.

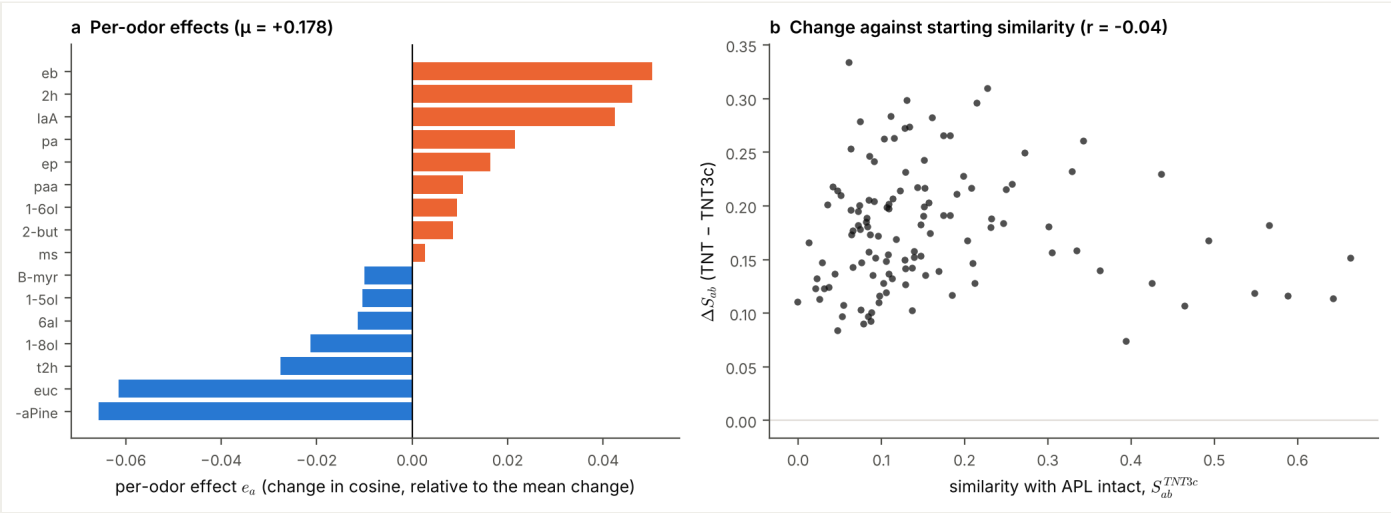


Figure 4. Which odors drive the change. a, The per-odor effect e_a of the additive fit, sorted. Orange: the odor's pairs rise more than average when APL is blocked. Blue: less. **b**, Each point is one odor pair: its similarity in TNT3c (horizontal) against its change (vertical).

4.1 What the per-odor effect means, mechanistically

The fit $\Delta S_{ab} \approx \mu + e_a + e_b$ says: the change of a pair is the average change, plus a bit that belongs to each of its two odors. e_a is how much more (or less) the pairs that contain odor a change than the

average pair. For example e_{eb} = +0.050: pairs with eb gain about 0.05 more cosine similarity than average when APL is blocked, whatever the partner. An odor with a large e_a has its whole row of the similarity matrix move up.

Mechanism: chance overlap. When an odor drives more units, it shares more responding units with every other odor just by chance: two unrelated 0/1 codes in which fractions p_a and p_b of equally excitable units respond have cosine $\sqrt{p_a p_b}$ (Report 08). So an odor that gains many responders when APL is blocked should gain similarity with all partners, which is exactly a large e_a. This can be tested on the data alone: compute the chance overlap $\sqrt{p_a p_b}$ from each recording's fractions responding, take its change TNT − TNT3c, fit the same additive model to it, and compare its per-odor effects with the real e_a (Figure 5b).

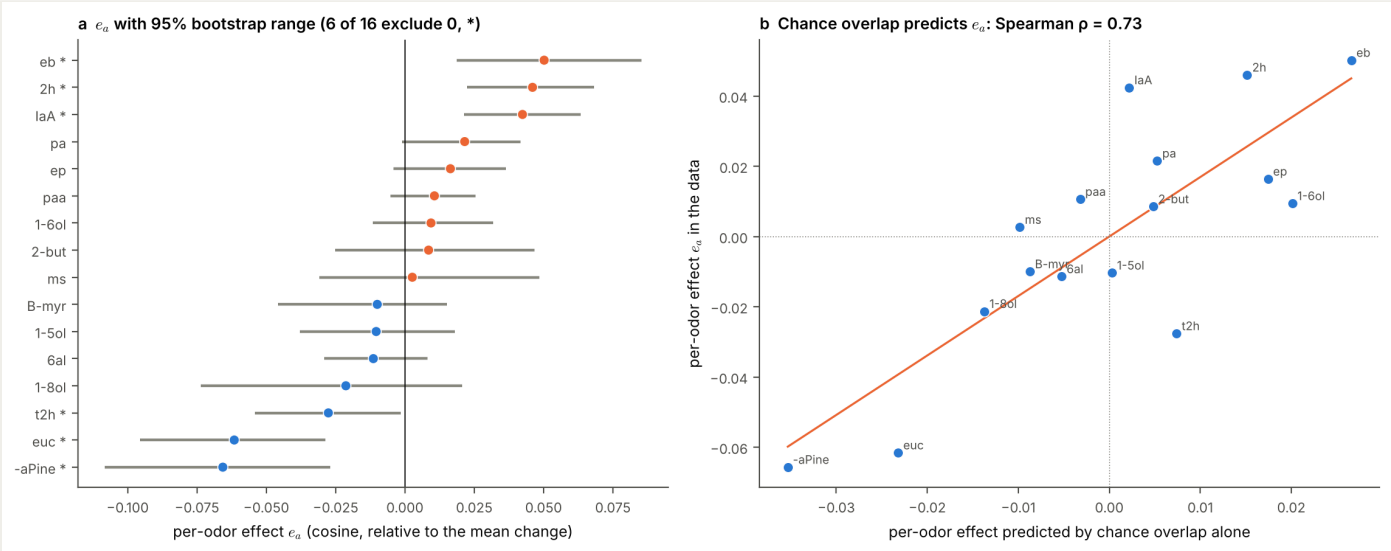


Figure 5. Per-odor effects: how sure, and why. a, The per-odor effects of Figure 4a, with their 95% bootstrap ranges (2,000 draws; recordings resampled with replacement within each condition; each draw refits the additive model). **b**, Each point is one odor. Horizontal: the per-odor effect of the additive fit to the change in chance overlap $\sqrt{p_a p_b}$ (fractions responding > 1, per recording, averaged). Vertical: the per-odor effect of the real change in similarity. The orange line is the least-squares fit.

The per-odor effects follow chance overlap (Spearman ρ = 0.73). They also follow how many units an odor drives in TNT (ρ = 0.76) and how many it gains (TNT − TNT3c, ρ = 0.73), but not the ratio TNT ÷ TNT3c (ρ = 0.23). So the per-odor part of the change is mostly the arithmetic of denser codes: odors that drive many units in TNT overlap more with everything. -aPine and euc gain the fewest responding units when APL is blocked (2.5 and 4.0 percentage points, against 5.7–13.9 for the other odors) and have the most negative effects. In mechanistic terms: if APL's feedback grows with the number of active KCs, as in the model of Report 05, it removes the most responders from odors that drive many KCs. Blocking it then adds the most responders to exactly those odors, and their codes overlap more with every partner. The data agree with the first step: the units an odor gains (TNT − TNT3c) rise with how many it drives in TNT3c (Spearman ρ = 0.82 over odors).

4.2 Are the per-odor effects real? The statistics

Table 3. Statistics for the per-odor effects. With 16 odors, the additive model has 16 free parameters for 120 pairs, so even random numbers give R^2 ≈ 15/119 ≈ 0.13. That is why the null is needed: 0.65 is far above it.

test	how	result

test	how	result
Is the additive fit better than chance?	Permutation: the 120 values of ΔS are shuffled across pairs (5,000 times), which removes any odor structure, and the additive model is refitted	R^2 = 0.65 in the data; null mean 0.13, 95th percentile 0.20; p = 0.0002 (none of 5,000 shuffles reach it)
Which e_a differ from 0?	95% bootstrap range over recordings (Figure 5a)	6 of 16 odors: -aPine, euc, t2h, IaA, 2h, eb
Are the e_a reproducible?	Split-half: two random halves of the recordings (4 + 5 and 4 + 6), e_a fitted in each, Pearson r over odors; 1,000 splits; Spearman–Brown to full data	r = 0.66 between halves, 0.80 for all the data

5 How reliable these numbers are

The recordings are the unit of resampling everywhere: bootstrap draws resample recordings with replacement within each condition, and split-half tests divide each condition's recordings into two random halves. A split-half correlation describes half the data. It is turned into an estimate for all the data with the Spearman–Brown formula, $r_{\text{full}} = 2r_{\text{half}}/(1 + r_{\text{half}})$.

- **Rank order of similarity.** Over the 120 pairs, S^{TNT} and S^{TNT3c} have ρ_s = 0.87. With half the recordings, two halves of the same condition give 0.90 (TNT3c) and 0.91 (TNT), while TNT vs TNT3c gives 0.83. The between-condition value is lower in 97.5% of splits. Dividing the between-condition value by the within-condition one gives about 0.92: the rank correlation expected without measurement noise. Report 09 and the local model use this as the data's value.
- **Change per pair.** The split-half reliability of ΔS is 0.70 (Spearman–Brown, all recordings), which caps any model's Pearson r with it near 0.84 (Report 07 explains why).
- **Duplicates.** No unit is repeated within or across recordings. Units whose nearest neighbour in their own recording is closer than expected are rare (0.5% of responsive TNT units, 1.6% TNT3c).
- **Two hemispheres of one fly.** Weighting each fly once instead of each recording changes the mean ΔS from 0.178 to 0.182, and the pattern over pairs correlates 0.99.

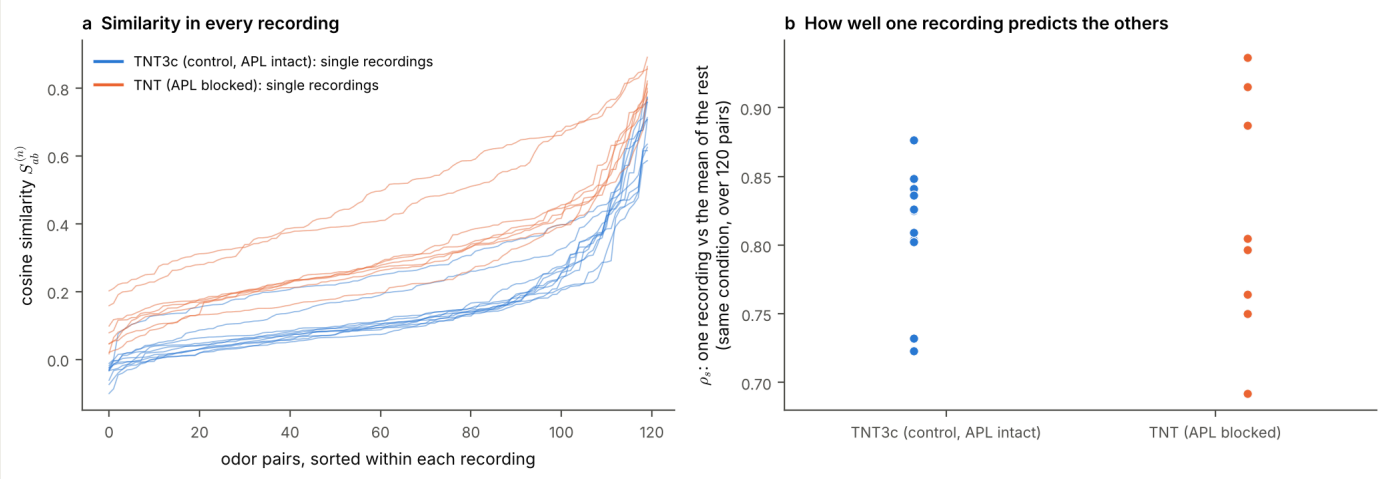


Figure 6. Consistency across recordings. a, Each line is one recording: its 120 pair similarities sorted from lowest to highest. TNT recordings (orange) sit higher throughout. **b**, Each point is one recording: the Spearman correlation of its pair similarities with the mean of the other recordings of the same condition. It measures how consistent the rank order of pairs is from hemisphere to hemisphere.

Caveats

The units are not matched across recordings or conditions, so every comparison is between condition means.

The responses are FC z-scores. Thresholding at 1 assumes that a z-score of 1 means the same in both genotypes.

Noise lowers cosine similarity more when responses are small. TNT3c has smaller responses, so part of the rise in TNT may be noise. Thresholding at 1 before the cosine gives the same picture (every pair still rises; pattern r 0.94 with the raw change).

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